

CSCI 4146 PROCESS OF DATA SCIENCE PROJECT-2

MODEL AUDIT MEMO: LENDINGCLUB DEFAULT RISK MODEL

Prepared for: Credit Risk Committee & Underwriting Operations
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1 Executive Summary

This audit evaluates the behavior and vulnerabilities of our gradient-boosted default prediction model trained on historical LendingClub records. The evaluation confirms that the model over-relies on FICO scores and interest rates as macro-proxies for creditworthiness, creating systematic blind spots. High-FICO, low-rate borrowers who default are consistently misclassified as safe (False Negatives), while mid-FICO applicants who have run routine credit checks are disproportionately flagged as risky (False Positives). Additionally, prediction errors are concentrated in high-volume, high-risk loan segments like debt consolidation. To address these vulnerabilities, we recommend a combined strategy: engineering ratio-based leverage metrics and applying sample weights to high-risk sub-populations during training.

2 Model Performance & Threshold Rationale

The baseline LightGBM model achieves a validation ROC-AUC of **0.6361**, showing that default prediction remains highly challenging.

We established the model's operational decision threshold at **0.3**. Operating at the default 0.5 threshold is economically unviable: it flags only 1.9% of loans as risky and yields an F1-score of 0.081, leaving the institution exposed to substantial default losses. Lowering the cutoff to 0.3 increases the validation F1-score to **0.2455**, capturing 19.87% of defaults (Recall) while maintaining a total loan rejection rate of 9.9% (Table 1). This risk-averse stance is financially justified because the loss of loan principal from a missed default far outweighs the interest revenue lost from turning away a creditworthy applicant.

Table 1: Validation Set Metrics Across Threshold Grids

Threshold	Validation F1	Recall	Positive Prediction (Rejection) Rate
0.3	0.2455	19.87%	9.92%
0.4	0.1231	7.17%	4.33%
0.5	0.0812	4.23%	1.98%
0.6	0.0438	2.28%	0.68%

3 Real-World Failure Modes (Local Explanations)

Using local explainability techniques (SHAP/LIME), we analyzed the individual credit decisions behind our validation errors, revealing clear patterns of systemic bias:

- **Case 7992 - False Positive (Predicted: 82.48% default risk, Actual: Paid in Full):** The model marked this borrower as highly risky, primarily driven by credit inquiries (`inq.last.6mths` SHAP contribution of +1.33) and high revolving utilization. Underwriters noted that while these indicate short-term credit-seeking, the borrower's stable debt-to-income (DTI) ratio made them viable. The model overreacted to these behavioral signals.
- **Case 3627 - False Negative (Predicted: 0.61% default risk, Actual: Defaulted):** The model predicted a near-zero probability of default due to the borrower's high FICO score (777) and low interest rate. The model used FICO as an absolute shield, completely ignoring high revolving credit balances that eventually drove the borrower to default.

4 Global Behaviors & Non-Linear Relationships

Our global audit confirms that the model relies heavily on macro-proxies instead of borrower cash flow:

- **Feature Importance:** Permutation testing identified the top global drivers as interest rate (`int.rate`, mean ROC-AUC drop of 0.0264), followed by recent inquiries (`inq.last.6mths`, 0.0203), monthly payments (`installment`, 0.0165), and FICO score (`fico`, 0.0158).
- **Risk Saturation:** Partial Dependence Plots (PDP) show that predicted risk remains flat until interest rates cross 11%, where it spikes sharply. Conversely, FICO score risk drops steadily but flattens out completely above 720–740. This flattening prevents the model from differentiating risk among high-FICO applicants.
- **Inquiry Double Standard:** We uncovered a strong interaction rule between FICO and inquiries. For excellent credit files ($FICO > 750$), inquiries are ignored. However, for mid-FICO applicants (660–700), the presence of 2+ inquiries triggers a disproportionate risk penalty, pushing them over the 0.3 threshold.

5 Subgroup Vulnerabilities & Risk Concentration

Slicing the validation set by loan purpose revealed significant performance weaknesses in high-stakes segments:

- **Debt Consolidation & Small Business:** Debt consolidation ($n = 802$) and small business ($n = 121$) represent the highest default rates (16.7% and 22.3%). However, the model exhibits its weakest performance on these segments, yielding poor ROC-AUCs of just **0.566** and **0.567** (Table 2).
- **Home Improvement:** By comparison, the model excels on home improvement loans ($n = 127$, default rate 15.7%), achieving an ROC-AUC of **0.768**.

Table 2: Validation Set Slicing Performance by Loan Purpose

Subgroup (Purpose)	Sample Size (n)	Observed Default Rate	Subgroup AUC	Subgroup F1	Recall
Major Purchase	86	10.47%	0.6118	0.1429	11.11%
Credit Card	246	12.20%	0.7338	0.1429	10.00%
Debt Consolidation	802	16.71%	0.5661	0.2300	17.16%
Small Business	121	22.31%	0.5670	0.3077	37.04%
Home Improvement	127	15.75%	0.7678	0.3750	30.00%

6 Remediation Strategy

To address these biases, we recommend a combined feature engineering and data-weighting approach:

1. **Feature Engineering:** Introduce leverage ratios like `dti_to_income` and `utilization_to_limit` to capture absolute debt burden, bypassing the model's over-reliance on static FICO values.
2. **Sample Weighting:** Apply higher sample weights to prediction errors in the debt consolidation and small business slices during model training, forcing the algorithm to learn segment-specific risks.

7 Limitations & Monitoring Plan

- **Limitations:** The audit is limited by the static nature of the LendingClub data and lacks macroeconomic indicators (such as inflation or unemployment rates) that heavily dictate default patterns.
- **Monitoring Plan:** We will run monthly Population Stability Index (PSI) checks on incoming FICO and interest rate distributions to track concept drift. Additionally, we will audit the risk threshold quarterly against actual defaults to maintain portfolio health.